



UNIVERSITY
OF COLOGNE

WORKSHOP 04

Advanced Analytics and Applications

Agenda

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Lecture Recap

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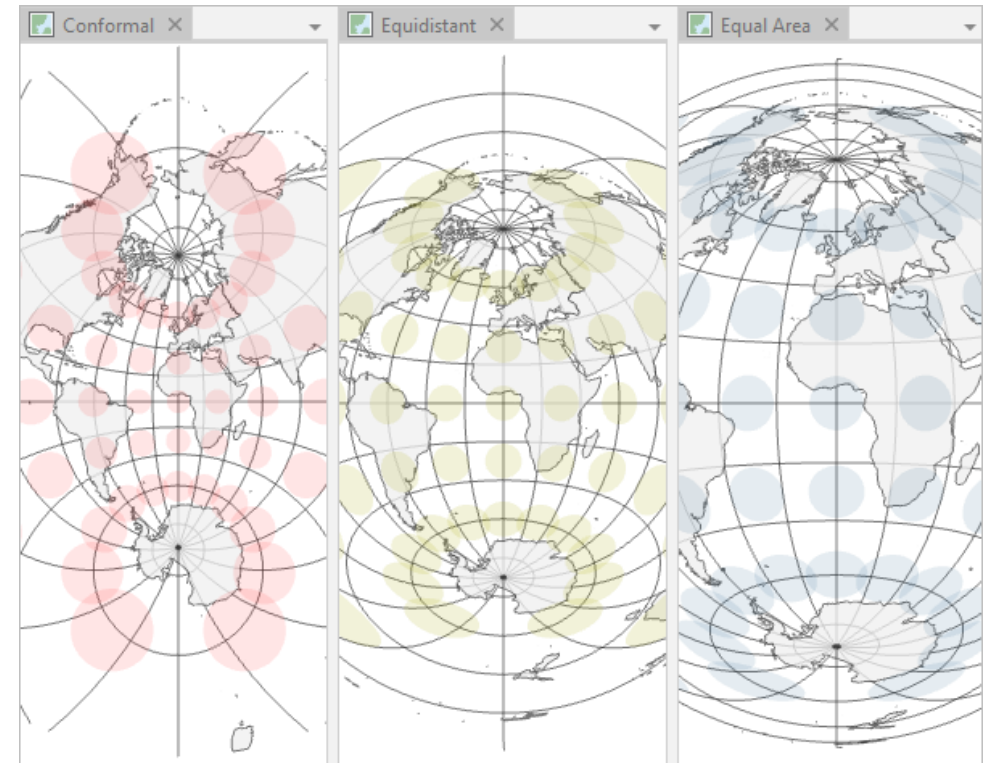
Exemplary Exam Questions

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Coding Tasks

Coordinate Reference Systems and Projections

- Coordinate Reference Systems (CRS) tells users what is where on earth using a coordinate system (like Lat/Long or X,Y,Z) and a reference surface (sphere, ellipsoid, flat map, ...)
- Projections between CRS. Most often map geographic CRS (3D) to projected 2D CRS with planar coordinates (X,Y).
- Because you cannot project without distortion, maps distort either shape, area distance, or direction (often even multiple).
- That means, projected maps are trade-offs on these projections, some work good for distances, others for area etc.



<https://learn.arcgis.com/en/projects/choose-the-right-projection/>

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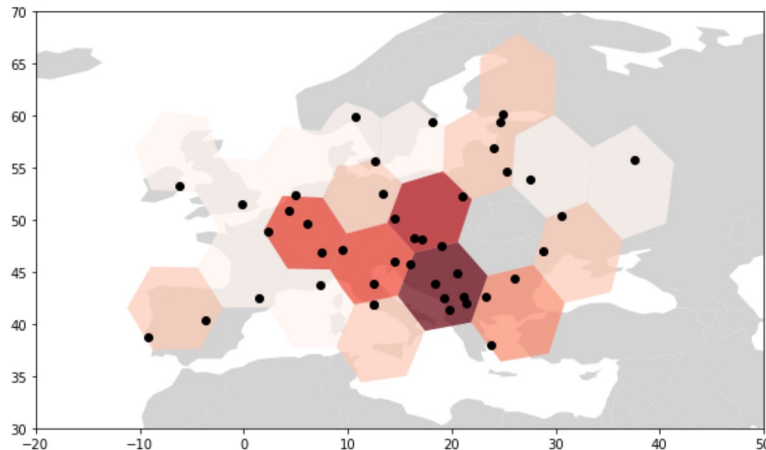
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Coding Tasks

Question 1: Python

Below, on the left, you see a map of European cities, as available in the data set, on a map with hexagonal cells that are colored in different shades of red. On the right you see incomplete Python code that produces the plot. The world data set ("**_lowres**") has country boundaries, the "**_cities**" data set has city coordinates.



```
1 # something is missing here
2 import h3
3 from shapely.geometry.polygon import Polygon
4
5 world = gpd.read_file(gpd.datasets.get_path("naturalearth_lowres"))
6 cities_path = gpd.datasets.get_path("naturalearth_cities")
7 # something is missing here, remember, only Europe!
8
9 cities['h3'] = [h3.geo_to_h3(c.x, c.y, resolution=1) for c in cities.geometry]
10 agg = cities.groupby('h3').count()['name']
11 agg = gpd.GeoDataFrame(agg).rename(columns={'name': 'cities'})
12 agg['geometry'] = [Polygon(h3.h3_to_geo_boundary(h, geo_json=False)) for h in agg.index]
13 agg.set_crs(epsg="4326", inplace=True)
14
15 fig, ax = plt.subplots(1,1, figsize=(10,10))
16
17 cities.plot(ax=ax, color='black', zorder=2)
18 # something is missing here
19 mask.plot(ax=ax, color='lightgrey', zorder=-1)
20
21 ax.set_ylim([30,70])
22 ax.set_xlim([-20,50])
23
24 plt.show()
```

Disclaimer: gpd.datasets is deprecated in newer versions. The workshop notebooks use the newer geodatasets.

Question 1: Python

Question:

- a) Please provide code for the missing lines 1, 7, and 10.
- b) If, in the plot, one would not aggregate over arbitrary cells but **countries** instead, a city would need to have its country as an attribute. When this attribute is not available, what do you need to do?

```
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```

Question 1: Python

Answer:

a) Line 1: `import geopandas as gpd`

Line 7: `cities = gpd.read_file(cities_path, mask=world[world.continent=='Europe'])`

Line 18: `agg.plot(ax=ax, alpha = 0.7, column='cities ', cmap='Reds')`

b) One would need to do a spatial join with the intersect/contains relationship test on the “world” data set to obtain the respective country for each city. After that, a choropleth map can be created using the same groupby and count technique.

Question 2: Visualization

Question: The local police created a map with total number of violent crimes committed per district. The press took it and printed headlines immediately “Lindenthal is dangerous! Chorweiler perfectly safe!”

- a) Can you trust the map? Why or why not?
- b) Explain a change you would suggest to implement

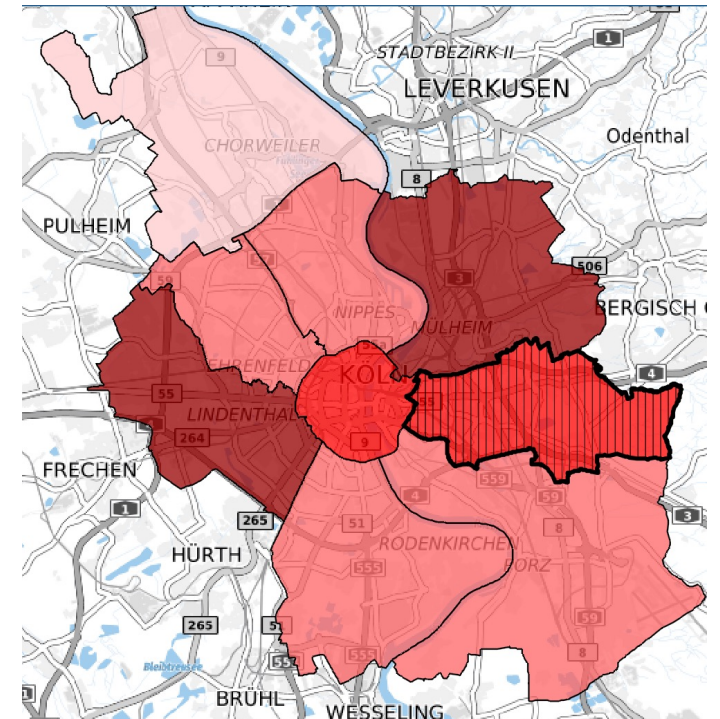


Figure 1. Some Map

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Answer:

- a) You cannot trust absolute numbers when they are not normalized for the number of inhabitants or the size of the district!
- b) You should rather look at rates normalized by e.g. inhabitants.
Alternatives: Point visualization / heatmap.

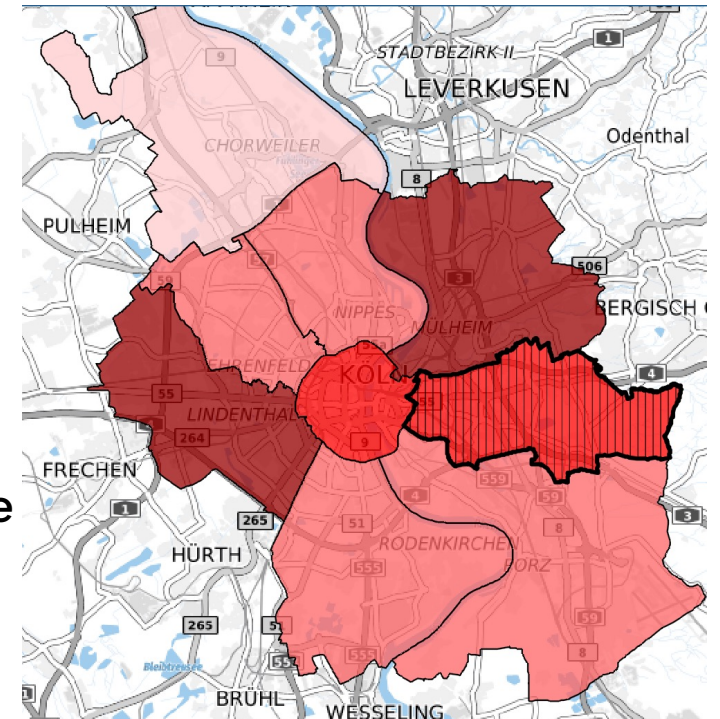


Figure 1. Number of inhabitants per district in Cologne

Multiple Choice Questions: Visualization

Question: When visualizing a spatial prediction model's output, you want to show not just the predicted distribution of values but also their associated prediction error. Which **methods** should be avoided?

- ☐ Using transparency based on uncertainty (error)
- ☐ Displaying standard errors as text labels on the map
- ☐ Reporting the average standard error below the map
- ☐ Creating two maps side by side, one with the prediction distribution, one with the error

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Bonus Question: City Characteristics and Biking

Question: How would you structure and carry out research to find out whether the characteristics of a city have an influence on how likely people will choose active transportation modes such as a bicycle for a trip from A to B?

Hints:

- Target Variable?
- Features? Which characteristics? Consider how easy it is to gain access to that data.
- Challenges?

Bonus Question: City Characteristics and Biking

Answer:

- **Target Variable:** Choice Probability (of biking vs other modes)
- **Features:** Geographical characteristics that are openly available (maps, geodata, sociodemographics)

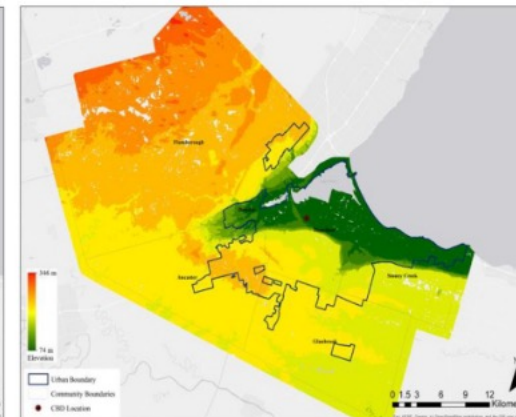
Read this paper if it interests you:

<https://doi.org/10.1016/j.jtrangeo.2021.103158>

It does exactly that and finds that improving built environment in cities make people choose walking and biking more often.



(a) Road network



(b) Elevation model

- **Challenges:** The prime challenge is to find a common spatial unit to (dis)aggregate all the variables to.

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Contact



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For specific enquiries regarding this course contact us by sending an email to the IS3 teaching address at is3-teaching@wiso.uni-koeln.de

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[AAA] <Your request subject>